Air, closedinthevesselatthetemperatureof 200ºC,exertsthe following pressure:

$$1\ atm \cdot \frac{473\ K}{293\ K} = 1{,}61\ atm$$

At the same temperature, the vapor pressure is$2{,}88 - 1{,}61 = 1{,}27\ atm$. Then, the vaporpressureatanytemperature$T$when all water is va orized is found as:

$$P_{vap} = 1{,}27 \cdot \frac{T}{473}\ atm = \frac{T}{373}\ atm$$

Let$P(T)$ be the temperature dependence of the saturated vapor pressure. Then, all water isevaporated at the temperature, which is derived from the following equation:

$$P(T) = \frac{T}{373}\ atm,$$

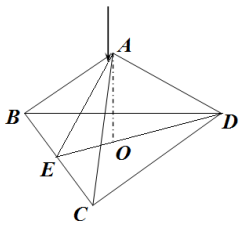
Solution of this equation could be obtained even without further knowledge of the function $P(T)$. It is well known, that at $P = 1\ atm, T = 373\ K = 100\ ℃$. Thus, this checks the solution.

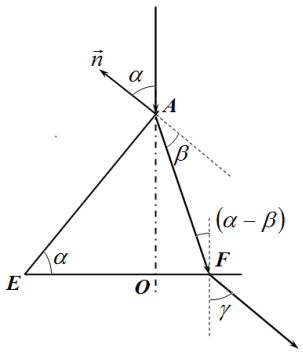
Marking scheme for Problem 1B:

№	Description	Points
1.	Correct value for air pressure at T=200℃	0,5
2.	Vapor pressure was found for any temperature T	0,5
3.	Conditionfor$T_{vap}$was written	0,5
4.	Correct final answer	0,5
Total		**2.0**

Problem 1.C Pyramid

1. Considera beam, which falls down on one of the sidesat the point, close to the top of the pyramid. The angle of incidence of the beam on the facet is equal to the angle between the lateral facet and the base of the pyramid $\angle AEO = \alpha$. It is easy to prove geometrically that $\cos\alpha = \frac{1}{\sqrt{3}}$, thus $\alpha = 54{,}7°$.

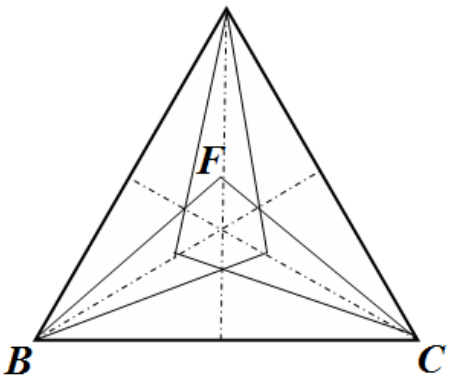




According tothe refraction law the angle β can be written as follows:

$\sin\beta = \frac{\sin\alpha}{n} = \frac{1}{n}\sqrt{\frac{2}{3}}$. Thus, $\beta = 24{,}6°$.

Let us find the point F, where that beam falls on the base of the pyramid. The height of the lateral facet is $|AE| = \frac{a}{2}$, pyramid's height $|AO| = \frac{a}{2}\sin\alpha = \frac{a}{2}\sqrt{\frac{2}{3}} = 0{,}41mm$.



Then, $|OF| = |AO|tg(\alpha-\beta) = 0{,}24\ mm$.Finally, the point sought is located at the distance $|DF| = |OD| - |OF| = \frac{a}{\sqrt{3}} - 0{,}24mm = 0{,}91mm$. The same point is located at a distance